# Compactification Independence, Lower-Shriek Base Change, and Weil Structures

## 1 Introduction

Let

$$f : X \longrightarrow Y$$

be a compactifiable morphism, for example a separated morphism of finite type between schemes in a setting where Nagata compactification is available.

The usual construction of the exceptional direct image

$$Rf_! : \mathcal{D}(X) \longrightarrow \mathcal{D}(Y)$$

starts by choosing a compactification

$$X \xrightarrow{j} \overline{X} \xrightarrow{p} Y,$$

where $j$ is an open immersion and $p$ is proper, and setting

$$Rf_{!,c} := Rp_* j_!,$$

where $c = (j, \overline{X}, p)$ denotes the chosen compactification.

The purpose of this note is to explain precisely what is required in order to make $Rf_!$ independent of the compactification, and how this gives a well-defined lower-shriek base-change isomorphism and hence a Weil structure on $Rf_!K$.

The main point is that no $\infty$-categorical formalism is required for this issue. Ordinary categories, functors, natural transformations, and coherent natural isomorphisms are sufficient.

## 2 The category of compactifications

Fix $f : X \to Y$. Let

$$\mathrm{Comp}(f)$$

denote the category of compactifications of $f$.

An object $c \in \mathrm{Comp}(f)$ is a factorization

$$c = \left( X \xrightarrow{j_c} \overline{X}_c \xrightarrow{p_c} Y \right)$$

such that $j_c$ is an open immersion, $p_c$ is proper, and

$$p_c j_c = f.$$

A morphism

$$u : c' \longrightarrow c$$

is a morphism

$$u : \overline{X}_{c'} \longrightarrow \overline{X}_c$$

over $Y$, restricting to the identity on $X$, such that

$$p_{c'} = p_c u$$

and

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{id}} & X \\ {\scriptstyle j_{c'}}\downarrow & & \downarrow{\scriptstyle j_c} \\ \overline{X}_{c'} & \xrightarrow[u]{} & \overline{X}_c \end{array}$$

is Cartesian.

One may choose the opposite convention for morphisms of compactifications. This changes whether the diagram below is indexed by $\mathrm{Comp}(f)$ or $\mathrm{Comp}(f)^{\mathrm{op}}$, but has no mathematical effect.

Under the usual hypotheses, $\mathrm{Comp}(f)$ is cofiltered. Concretely:

1. it is nonempty;
2. any two compactifications admit a common refinement;
3. any two parallel morphisms of compactifications become equal after passing to a further refinement.

The third condition is important: it is what eventually makes the comparison isomorphism independent of the choice of common refinement.

## 3 The compactification diagram

For every compactification $c$, define

$$F_c := Rp_{c*}j_{c!}.$$

Thus

$$F_c : \mathcal{D}(X) \longrightarrow \mathcal{D}(Y).$$

Suppose

$$u : c' \longrightarrow c$$

is a morphism of compactifications. Since $u$ is the identity over $X$, there is a canonical natural isomorphism

$$Ru_* j_{c'!} \xrightarrow{\sim} j_{c!}.$$

Consequently,

$$\begin{aligned} F_{c'} &= Rp_{c'*}j_{c'!} \\ &\simeq Rp_{c*}Ru_*j_{c'!} \\ &\simeq Rp_{c*}j_{c!} \\ &= F_c. \end{aligned}$$

Denote the resulting natural isomorphism by

$$\theta_u : F_{c'} \xrightarrow{\sim} F_c.$$

The crucial point is not merely that $\theta_u$ is an isomorphism. One needs the family $\theta_u$ to satisfy

$$\theta_{\mathrm{id}_c} = \mathrm{id}_{F_c}$$

and

$$\theta_{u\circ v} = \theta_u \circ \theta_v$$

for composable refinement morphisms

$$c'' \xrightarrow{v} c' \xrightarrow{u} c.$$

Thus the compactifications determine an ordinary categorical diagram

$$F : \mathrm{Comp}(f) \longrightarrow \mathrm{Fun}\big(\mathcal{D}(X), \mathcal{D}(Y)\big)$$

which sends every morphism in $\mathrm{Comp}(f)$ to an isomorphism.

If one uses the opposite convention for refinement morphisms, the same construction is written

$$F : \mathrm{Comp}(f)^{\mathrm{op}} \longrightarrow \mathrm{Fun}\big(\mathcal{D}(X), \mathcal{D}(Y)\big).$$

Nothing in this construction requires an $\infty$-category.

# 4 Canonical comparison between two compactifications

The following elementary categorical observation is the key point.

**Lemma 1.** *Let $I$ be a cofiltered category and*

$$F : I \longrightarrow \mathcal{C}$$

*a functor which sends every morphism of $I$ to an isomorphism. Then for every pair $i, j \in I$, there is a canonical isomorphism*

$$\kappa_{i,j} : F(i) \xrightarrow{\sim} F(j)$$

*characterized by compatibility with the transition morphisms. Moreover,*

$$\kappa_{i,i} = \mathrm{id}, \qquad \kappa_{j,k} \circ \kappa_{i,j} = \kappa_{i,k}.$$

*Proof.* Choose an object $r \in I$ and morphisms

$$a : r \to i, \qquad b : r \to j.$$

Define

$$\kappa_{i,j} := F(b) \circ F(a)^{-1}.$$

We must show that this is independent of the choice of $r, a, b$.

Suppose another choice is

$$a' : r' \to i, \qquad b' : r' \to j.$$

By cofilteredness, there is an object mapping to both $r$ and $r'$. After a further refinement, the two induced maps to $i$ can be made equal, and similarly the two induced maps to $j$ can be made equal.

Applying $F$, and using the fact that every transition morphism is an isomorphism, one obtains

$$F(b)F(a)^{-1} = F(b')F(a')^{-1}.$$

Thus $\kappa_{i,j}$ is independent of all choices.

The identities

$$\kappa_{i,i} = \mathrm{id}$$

and

$$\kappa_{j,k}\kappa_{i,j} = \kappa_{i,k}$$

follow by choosing a common refinement of the relevant finite collection of objects. □

Applying this lemma to the compactification diagram gives canonical natural isomorphisms

$$\kappa_{c,d} : F_c \xrightarrow{\sim} F_d$$

for all $c, d \in \mathrm{Comp}(f)$, satisfying

$$\kappa_{c,c} = \mathrm{id}$$

and

$$\kappa_{d,e} \circ \kappa_{c,d} = \kappa_{c,e}.$$

Furthermore, if

$$u : c' \to c$$

is itself a refinement morphism, then

$$\kappa_{c',c} = \theta_u.$$

This is the precise ordinary-categorical meaning of the statement that $Rf_!$ is independent of the choice of compactification.

## 5 What "canonical" means

It is important to distinguish the following two statements.

The weak statement is

$$F_c \simeq F_d$$

for every pair of compactifications $c, d$.

This is not sufficient for many constructions. There may be many different isomorphisms

$$F_c \xrightarrow{\sim} F_d.$$

If

$$\alpha : F_c \xrightarrow{\sim} F_d$$

is one such isomorphism and

$$a : F_d \xrightarrow{\sim} F_d$$

is a nontrivial automorphism, then

$$a \circ \alpha$$

is another.

What is actually required is a specified family

$$\kappa_{c,d} : F_c \xrightarrow{\sim} F_d$$

which satisfies the cocycle relations

$$\kappa_{c,c} = \mathrm{id}, \qquad \kappa_{d,e}\kappa_{c,d} = \kappa_{c,e},$$

and which agrees with the geometrically defined transition morphisms along refinements.

Thus "canonical" here means

unique after requiring compatibility with the entire refinement system.

This is an entirely 1-categorical statement.

## 6 Defining $Rf_!$ using one chosen compactification

For formalization, there is no need to define $Rf_!$ as a literal colimit over all compactifications.

Choose once and for all a compactification

$$c_f \in \mathrm{Comp}(f)$$

and define

$$Rf_! := F_{c_f}.$$

For every other compactification $c$, one then has the canonical comparison

$$\kappa_{c_f,c} : Rf_! \xrightarrow{\sim} F_c.$$

The choice of $c_f$ is therefore merely an implementation choice. All mathematically relevant constructions should use the canonical comparison system $\kappa$.

In a formal proof assistant, the compactification $c_f$ may be chosen noncomputably. This causes no mathematical problem, provided that the resulting API exposes the canonical comparison maps and their coherence properties.

## 7 Lower-shriek base change

Consider a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y. \end{array}$$

Let

$$c = \big(X \hookrightarrow \overline{X} \to Y\big)$$

be a compactification of $f$.

Base change of $c$ along $g$ gives a compactification

$$g^*c$$

of $f'$:

$$X' \hookrightarrow \overline{X} \times_Y Y' \longrightarrow Y'.$$

At the level of this particular compactification, proper base change and base change for extension by zero produce a canonical natural isomorphism

$$\beta_c : g^*F_c \xrightarrow{\sim} F_{g^*c}h^*.$$

The essential compatibility statement is that $\beta_c$ is natural with respect to refinement of compactifications.

For two compactifications $c, d$, one requires

$$\big(\kappa_{g^*c,g^*d}h^*\big) \circ \beta_c = \beta_d \circ \big(g^*\kappa_{c,d}\big).$$

Equivalently, the diagram

$$\begin{array}{ccc} g^*F_c & \xrightarrow{\beta_c} & F_{g^*c}h^* \\ {\scriptstyle g^*\kappa_{c,d}}\downarrow & & \downarrow{\scriptstyle \kappa_{g^*c,g^*d}h^*} \\ g^*F_d & \xrightarrow[\beta_d]{} & F_{g^*d}h^* \end{array}$$

commutes.

This is the compactification-independence statement for lower-shriek base change.

# 8 The global base-change isomorphism

Suppose we have chosen compactifications

$$c_f \in \mathrm{Comp}(f), \qquad c_{f'} \in \mathrm{Comp}(f').$$

By definition,

$$Rf_! = F_{c_f}, \qquad Rf'_! = F_{c_{f'}}.$$

The compactification-level base-change map gives

$$g^*F_{c_f} \xrightarrow{\beta_{c_f}} F_{g^*c_f}h^*.$$

The target uses the compactification $g^*c_f$, whereas our chosen definition of $Rf'_!$ uses $c_{f'}$.

This is exactly where compactification independence is needed.

Use the canonical comparison

$$\kappa_{g^*c_f,c_{f'}} : F_{g^*c_f} \xrightarrow{\sim} F_{c_{f'}}.$$

Define

$$\mathrm{BC}_! : g^*Rf_! \xrightarrow{\sim} Rf'_!h^*$$

by

$$\boxed{\mathrm{BC}_! = \big(\kappa_{g^*c_f,c_{f'}}h^*\big) \circ \beta_{c_f}.}$$

Thus

$$g^*Rf_! = g^*F_{c_f} \xrightarrow{\beta_{c_f}} F_{g^*c_f}h^* \xrightarrow{\kappa_{g^*c_f,c_{f'}}h^*} F_{c_{f'}}h^* = Rf'_!h^*.$$

The compatibility of $\beta_c$ with $\kappa_{c,d}$ implies that this construction does not depend on replacing the chosen compactifications by other ones.

Consequently, lower-shriek base change is not merely the assertion that the two functors

$$g^*Rf_! \qquad \text{and} \qquad Rf'_!h^*$$

happen to be isomorphic.

It consists of a specified natural isomorphism

$$\mathrm{BC}_! : g^*Rf_! \xrightarrow{\sim} Rf'_!h^*$$

constructed from the compactification-level base-change maps and the canonical compactification-comparison maps.

## 9 Application to Weil structures

Let $X$ and $Y$ be defined over a finite field, and suppose

$$f : X \to Y$$

is defined over that field.

Let

$$\mathrm{Frob}_X : X \to X, \qquad \mathrm{Frob}_Y : Y \to Y$$

denote the chosen Frobenius morphisms.

Assume that the square

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{Frob}_X} & X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[\mathrm{Frob}_Y]{} & Y \end{array}$$

is the relevant Cartesian square for the base-change formalism.

Let $K \in \mathcal{D}(X)$ carry a Weil structure

$$\varphi_K : \mathrm{Frob}_X^* K \xrightarrow{\sim} K.$$

Lower-shriek base change gives the specified isomorphism

$$\mathrm{BC}_! : \mathrm{Frob}_Y^* \, Rf_! \xrightarrow{\sim} Rf_! \, \mathrm{Frob}_X^* \, .$$

We may therefore define

$$\varphi_{Rf_!K} : \mathrm{Frob}_Y^* \, Rf_!K \xrightarrow{\sim} Rf_!K$$

by the composite

$$\boxed{\varphi_{Rf_!K} = Rf_!(\varphi_K) \circ \mathrm{BC}_{!,K}.}$$

Explicitly,

$$\mathrm{Frob}_Y^* \, Rf_!K \xrightarrow{\mathrm{BC}_{!,K}} Rf_! \, \mathrm{Frob}_X^* \, K \xrightarrow{Rf_!(\varphi_K)} Rf_!K.$$

This is a genuine piece of structure, not merely a proposition asserting the existence of some isomorphism.

## 10 Why the existence of an isomorphism is not enough

Suppose one knows only that

$$\mathrm{Frob}_Y^* \, Rf_!K \simeq Rf_! \, \mathrm{Frob}_X^* \, K.$$

This does not determine a Weil structure.

Indeed, if

$$\alpha : \mathrm{Frob}_Y^* \, Rf_!K \xrightarrow{\sim} Rf_! \, \mathrm{Frob}_X^* \, K$$

is one isomorphism and

$$a : Rf_! \, \mathrm{Frob}_X^* \, K \xrightarrow{\sim} Rf_! \, \mathrm{Frob}_X^* \, K$$

is an automorphism, then

$$a \circ \alpha$$

is another isomorphism.

After composing with $Rf_!(\varphi_K)$, these two choices may give different Weil structures on $Rf_!K$.

Therefore the proposition

$$\exists\, \alpha, \quad \alpha \text{ is an isomorphism}$$

is not the correct datum.

One needs an actual specified morphism

$$\alpha$$

together with a proof that it is an isomorphism.

In a formalization, there is an important distinction between

$$\text{“a specified morphism } \alpha \text{ with IsIso}(\alpha)\text{”}$$

and

$$\text{“there merely exists some isomorphism.”}$$

The former is enough; the latter is not canonical.

## 11 What is needed in a Lean formalization

A convenient organization is the following.

For each compactification $c$, define

$$\text{rfShriekAt}(c) := F_c.$$

For every refinement

$$u : c' \to c,$$

construct a specified natural transformation

$$\text{transition}(u) : F_{c'} \longrightarrow F_c$$

and prove that it is an isomorphism.

Then prove

$$\text{transition}(\text{id}) = \text{id}$$

and

$$\text{transition}(u \circ v) = \text{transition}(u) \circ \text{transition}(v).$$

From cofilteredness of $\text{Comp}(f)$, define

$$\text{compactificationIso}(c, d) : F_c \xrightarrow{\sim} F_d$$

and prove

$$\text{compactificationIso}(c, c) = \text{id}$$

and

$$\text{compactificationIso}(d, e) \circ \text{compactificationIso}(c, d) = \text{compactificationIso}(c, e).$$

One may then choose a preferred compactification $c_f$ and define

$$Rf_! := F_{c_f}.$$

The lower-shriek base-change isomorphism is defined by composing

compactification-level base change

with

the canonical compactification comparison.

Finally, the Weil structure on $Rf_!K$ is defined using this specified base-change isomorphism.

## 12 Do we need an $\infty$-category?

For the problem discussed above, the answer is no.

The required data are:

1. categories of compactifications;
2. functors between ordinary derived categories;
3. natural transformations;
4. natural isomorphisms;
5. equalities expressing compatibility under composition.

This is ordinary 1-categorical and, when one keeps track of natural transformations between functors, 2-categorical coherence.

One may phrase the construction as a functor

$$\mathrm{Comp}(f) \longrightarrow \mathrm{Fun}\big(\mathcal{D}(X), \mathcal{D}(Y)\big)$$

whose transition morphisms are all isomorphisms.

Cofilteredness of $\mathrm{Comp}(f)$ then produces the coherent canonical comparison maps.

An $\infty$-categorical enhancement becomes useful if one wants to construct the entire six-functor formalism together with all higher coherences at once, or if one wishes to retain derived mapping spaces rather than only the homotopy categories.

However, none of those higher coherences is needed merely to solve the compactification-independence problem required for defining lower-shriek base change and the induced Weil structure.

## 13 Summary

The essential structure is

$$\boxed{\mathrm{Comp}(f) \longrightarrow \mathrm{Fun}\big(\mathcal{D}(X), \mathcal{D}(Y)\big), \qquad c \longmapsto Rp_{c*}j_{c!},}$$

with every refinement sent to a specified natural isomorphism.

Cofilteredness of $\mathrm{Comp}(f)$ then gives canonical comparison isomorphisms

$$\kappa_{c,d} : Rf_{!,c} \xrightarrow{\sim} Rf_{!,d}$$

satisfying the cocycle identity

$$\kappa_{d,e}\kappa_{c,d} = \kappa_{c,e}.$$

The lower-shriek base-change map is obtained by

$$g^* Rf_{!,c_f} \xrightarrow{\beta_{c_f}} Rf_{!,g^* c_f} h^* \xrightarrow{\kappa} Rf'_{!,c_{f'}} h^*.$$

This specified base-change isomorphism then defines the Weil structure

$$\mathrm{Frob}_Y^* \, Rf_! K \longrightarrow Rf_! \, \mathrm{Frob}_X^* \, K \longrightarrow Rf_! K.$$

Thus the relevant issue is not higher category theory, but rather the construction of a coherent system of compactification comparison isomorphisms in ordinary category theory.